Pranay Kapse

CAD Automation Engineer

Design Automation | Autodesk Inventor | iLogic | Inventor API | VB.NET | C# | VBA

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Portfolio: pranaykapse1.github.io/portfolio
Male | Date of birth: 29 January 1996 | Married



PROFESSIONAL SUMMARY

CAD Automation Engineer with 9+ years of experience in mechanical design and design automation for industrial and process equipment. Expert in Autodesk Inventor (3D, advanced parametric and skeleton modelling, iLogic, VBA, detailing), with programming in iLogic, VB.NET, VBA, C# and the Inventor API. I build Excel-driven product configurators and automate 3D models, assemblies, drawings and BOMs, helping organisations automate design and drafting workflows and standardise CAD workflows to reduce drafting time. I understand PLM and PDM, coordinate with team members and customers on technical requirements through to planning, scheduling, execution and documentation, and have worked with global teams in Denmark, the USA, China and Malaysia, including several visits to Denmark for standardisation projects.

CORE COMPETENCIES

- **CAD Automation:** *Autodesk Inventor API, i-Logic programming, VB.NET, VBA, C#, scripting and macros, Excel automation, WinForms user interfaces, design automation, product configurators, rule-based and parameter-driven design, Customization, CAD templates and masters.*
- **Design and Drafting Automation:** *3D Modelling, drawing automation, BOM automation, assembly generation, skeleton-driven modelling, top-down design, standardisation of CAD workflows, Content Center, Autodesk Vault*
- **CAD Software:** *Autodesk Inventor (expert), AutoCAD, SolidWorks, Siemens NX, Creo.*
- **Mechanical Design:** *pressure vessels, vacuum heat exchangers, plate heat exchangers (PHE), evaporation plants, tanks, skids and frames, ducting, piping and equipment layout, P&ID, fabrication and general arrangement drawings, GD&T, sheet metal, weldments, large assemblies, animation and rendering.*
- **PDM/PLM:** *Autodesk Vault, Autodesk Fusion Lifecycle 360 (PLM), MOJO, Movex.*
- **Standards:** *ASME, ANSI, DIN, EN, ASTM.*
- **Tools:** *MS Office (Word, Excel), Microsoft Teams, AI tools (ChatGPT, Claude) and API models for faster scripting and documentation.*
- **Strengths:** *creative Innovative problem solutions, team player, works independently, quick learner new technologies.*

PROFESSIONAL EXPERIENCE

Senior Engineer – CAD Automation | Kelvion Pvt. Ltd., India
Sep 2026 – Present

- *Develop CAD automation solutions for plate heat exchangers (PHE), skids and related equipment, automating repetitive design and drafting workflows.*
- *Standardise CAD workflows and modelling practices so standard configurations can be produced faster and more consistently.*

CAD Automation / Design Engineer | Alfa Laval India Pvt. Ltd., Pune, India
Jun 2020 – Sep 2026

Domain: oil and gas and process equipment. Software: Autodesk Inventor, iLogic, Inventor API, VB.NET, VBA, C#, Excel, AutoCAD, Autodesk Vault, Movex

- *Designed and maintained Autodesk Inventor automation, using iLogic rules, VB.NET and VBA script programing and the Inventor API, that automated repetitive design and drafting workflows for the engineering team.*
- *Built a fully automated* **Evaporation Plant Configurator** *in Inventor and iLogic, driven by an Excel input sheet. It generates tanks, frames, skids, ducting and plate heat exchangers, positions components and subassemblies, applies pipe standards and produces the related drawings.*
- *Reduced repetitive modelling and drafting effort and drafting errors, gave standard plant configurations a consistent output, and made the workflow easier for new engineers to use.*
- *Created WinForm user interfaces and Excel automation so engineers could enter inputs and run automation rules without editing code.*
- *Automated BOM generation & drawing creation, and standardised CAD workflows & modelling practices for the team.*

- *Modelled oil and gas pressure vessels, vacuum heat exchangers, skids and frames using iLogic and skeleton modelling, and supported new product development of vacuum heat exchangers; produced 3D plant layouts and foundation drawings, assemblies and drafting.*
- *Created animations and renderings for design reviews.*
- *Coordinated with team members and customers to understand technical requirements, with project planning, scheduling, execution and documentation.*
- *Worked with teams in Denmark, the USA, China and Malaysia; visited Denmark several times for standardisation projects and promoted global adoption of automation practices.*
- *Provided technical support, new features and documentation for existing designs and automation tools.*

Junior Design Engineer | **Teknovance Solution, India** *Mar 2018 – May 2020*

- *Created 3D models and 2D drawings of casting, forging and machined components (regulators, relief valves), oil and gas pressure vessels, structures, sheet-metal parts and weldments, using parametric modelling, iParts and iLogic in Autodesk Inventor, AutoCAD, SolidWorks and Siemens NX 12.*
- *Produced fabrication and general arrangement drawings with welding information and BOMs to GD&T and ASME standards; designed ladders, platforms and electrical panel enclosures.*
- *Worked from PLM project allocations with inputs and releases managed in Vault (Fusion Lifecycle 360, MOJO); produced animations and renderings.*
- *Handled projects independently across different clients.*

EDUCATION

- **Bachelor of Engineering (B.E.) Mechanical Engineering** – *Sinhgad Inst, Lonavala, Pune University (2013 – 2017)*
- **HSC** – *J.M. Patel College, Bhandara, Maharashtra Board (2013)*
- **SSC** – *Govind Vidhalaya, Palandur, Maharashtra Board (2011)*

ADDITIONAL INFORMATION

- **Projects:** *designed and built a fatigue testing machine (B.E. final-year project); designed and manufactured a petrol-powered go-kart (AIR 21 in competition); lab innovation on a four-bar mechanism*
- **Internships:** *Ford service centre; vehicle engines*
- *GATE 2018 qualified.*
- **Training:** *AutoCAD, Creo Parametric, CATIA V5R20, Autodesk Inventor, PLC; workshops on go-kart design, Hypermesh and robotics (quadcopter, automated robot)*
- **Areas of interest:** *3D parametric modelling, customisation, CAD automation and modules, research and development, product development.*